



SOC Practical Application Report

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Introduction

This report documents the practical implementation of Security Operations Center (SOC) monitoring activities conducted within a controlled lab environment. The objective of this exercise was to simulate real-world security monitoring scenarios, including brute-force detection, log analysis, browser artifact investigation, dashboard creation, and alert rule configuration.

The lab environment was built using Windows and Ubuntu virtual machines integrated with SIEM platforms. The exercises demonstrate how SOC analysts detect, investigate, and document suspicious authentication attempts and system-level activities using structured workflows and standardized monitoring tools.

This documentation serves as academic evidence of practical SOC capability and understanding of detection engineering concepts.

1. Log Analysis Practice

1.1 Windows Event Log Monitoring

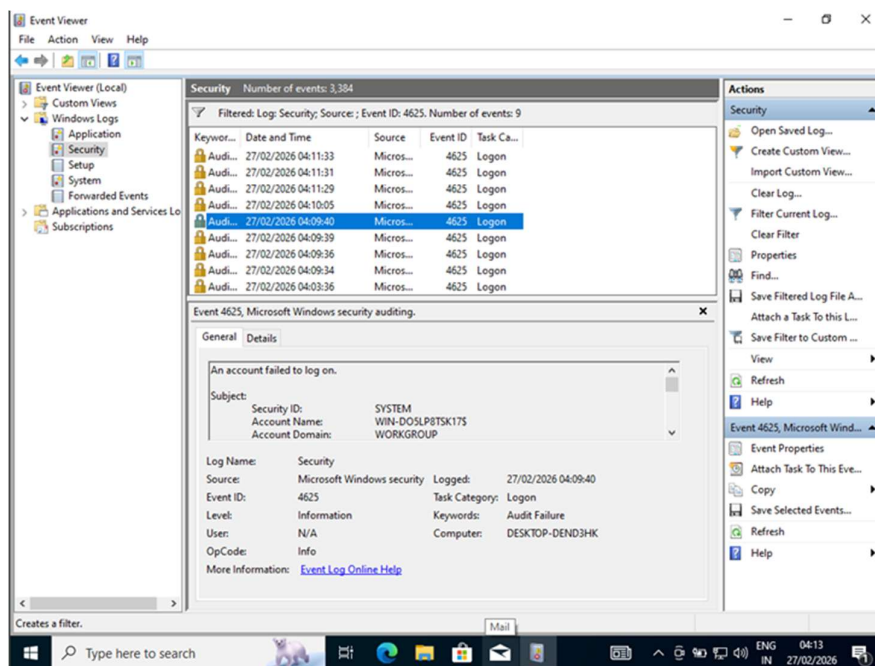
Windows Event Viewer was used to monitor security-related logs.

The following Event IDs were analyzed:

- Event ID 4625 – Failed Login Attempt
- Event ID 7045 – New Service Creation

Procedure:

- Open Event Viewer
- Navigate to: Windows Logs → Security
- Apply filter for Event ID 4625
- Review login failure patterns



Observations:

Multiple failed log in attempts from the same source IP within a short time period were identified. This pattern indicates a potential brute-force authentication attack.

Event ID 7045 was also reviewed to detect unauthorized service installations that may indicate persistence mechanisms.

1.2 Browser History Analysis

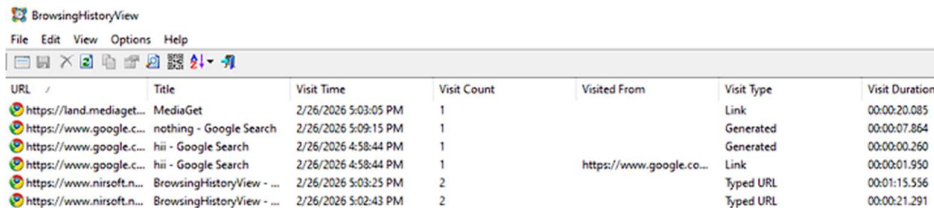
Browser artifacts were analyzed using Eric Zimmerman's forensic tool (LECmd).

Objective:

To detect visits to a test URL

Procedure:

- Located Chrome history database file
- Parsed file using LECmd
- Extracted visited URLs and timestamps



URL	Title	Visit Time	Visit Count	Visited From	Visit Type	Visit Duration
https://land.mediaget...	MediaGet	2/26/2026 5:03:05 PM	1		Link	00:00:20.085
https://www.google.c...	nothing - Google Search	2/26/2026 5:09:15 PM	1		Generated	00:00:07.864
https://www.google.c...	hii - Google Search	2/26/2026 4:58:44 PM	1		Generated	00:00:00.260
https://www.google.c...	hii - Google Search	2/26/2026 4:58:44 PM	1	https://www.google.co...	Link	00:00:01.950
https://www.nirsoft.n...	BrowsingHistoryView - ...	2/26/2026 5:03:25 PM	2		Typed URL	00:01:15.556
https://www.nirsoft.n...	BrowsingHistoryView - ...	2/26/2026 5:02:43 PM	2		Typed URL	00:00:21.291

Outcome:

The test URL was successfully identified in the browser history output, demonstrating forensic investigation capability within endpoint monitoring scenarios.

1.3 Advanced Task: Brute Force Detection

Failed login attempts were generated in a Windows Virtual Machine using repeated incorrect password entries.

The security logs were:

1. Filtered using Event Viewer for Event ID 4625
2. Exported into CSV format
3. Analyzed for brute-force attack indicators

Steps Performed:

- Generated repeated login failures
- Filtered logs using Event ID 4625
- Observed timestamps and source IP
- Exported filtered logs into CSV format



	A	B	C	D	E	F	G	H	I
1	Keywords	Date and Time	Source	Event ID	Task Category	PER ACCOUNT: FAILED TO LOG ON.			
						Subject: Security ID:SYSTEM Account Name:EDITH\$ Account Domain:WORKGROUP Logon ID:0x3E7 Logon Type:2 Account For Which Logon Failed: Security ID:NULL SID Account Name:- Account Domain:- Failure Information: Failure Reason:An Error occurred during Logon. Status:0xC000006D Sub Status:0xC0000380 Process Information: Caller Process ID:0xb50 Caller Process Name:C:\Windows\System32\svchost.exe Network Information: Workstation Name:- Source Network Address:127.0.0.1 Source Port:0			
2	Audit Failu	26-02-2026 16:49	Microsoft-Windows-Security-Auditing	4625	Logon	PER ACCOUNT: FAILED TO LOG ON.			

Result:

The exported logs clearly showed multiple authentication failures originating from the same IP address within a short time frame, confirming brute-force behavior.

2. Security Event Documentation

Security events were recorded using a structured SOC documentation template to maintain consistency in log tracking and incident analysis.

Security events were documented using the following template:

Date	Source ip	Event ID	Description	Action Taken
27-02-26	192.168.1.9	4625	Multiple Failed Login Attempts	Alert Reviewed & Logged

Importance of Documentation:

- Maintains investigation timeline
- Ensures audit compliance
- Supports incident response tracking
- Assists in future forensic review



3. Monitoring Dashboard Setup

To visualize and analyze security events effectively, monitoring dashboards were created using Kibana.

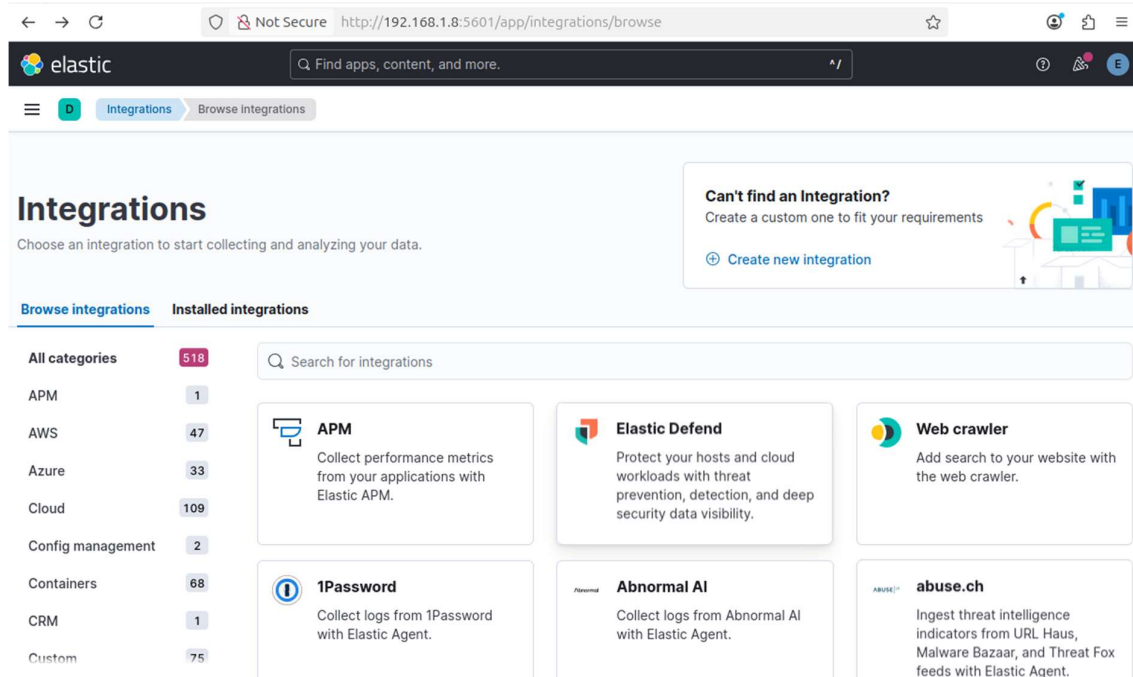
Steps Performed:

- Configured Kibana dashboards to monitor real-time security events
- Created visualizations for:
 1. Top 10 Source IPs Generating Alerts – helps identify suspicious or malicious IP activity
 2. Frequency of Critical Event IDs – highlights repeated security incidents such as failed logins
 3. Integrated pre-built detection rules (e.g., Sigma rules) to enhance visibility

Outcome:

- Enabled centralized monitoring of logs
- Improved detection of abnormal patterns and attack attempts
- Simplified analysis using graphical representations

```
ubuntu@ubuntu-VirtualBox:~$ sudo systemctl status kibana
[sudo] password for ubuntu:
● kibana.service - Kibana
   Loaded: loaded (/usr/lib/systemd/system/kibana.service; disabled; preset: enabled)
   Active: active (running) since Thu 2026-03-26 15:26:46 IST; 43min ago
     Docs: https://www.elastic.co
   Main PID: 13690 (node)
    Tasks: 11 (limit: 12560)
  Memory: 728.5M (peak: 1.2G swap: 17.1M swap peak: 18.4M)
     CPU: 2min 56.056s
    CGroup: /system.slice/kibana.service
```



4. Alert Rule Configuration

Custom alert rules were implemented in Elastic SIEM and Wazuh to detect suspicious activities.

Rule 1: Failed Login Detection (Elastic SIEM)

Rule Name: Detect Multiple Failed Logins

- Index: security-login-*
- Condition: Trigger alert when count > 5 failed logins within 5 minutes

Testing:

- Simulated failed SSH login attempts using incorrect credentials
- Verified alert generation in the SIEM dashboard

Tools Used:

- Wazuh – SIEM and threat detection
- Osquery – Endpoint monitoring and visibility
- Kibana – Data visualization and dashboards

Methodology:

- Built a mini SOC lab environment to simulate real-world scenarios
- Collected and analyzed logs from endpoints
- Mapped detected activities to MITRE ATT&CK framework



- Example: PowerShell-based attacks mapped to T1059 (Command Execution)

Outcome:

- Gained practical experience in:
- Security monitoring
- Alert creation and tuning
- Threat detection and analysis

4.1 Advanced Task: SSH Brute Force Detection using Wazuh SIEM

A simulated SSH brute-force attack was performed from a Windows host to an Ubuntu server using invalid login credentials.

The Wazuh SIEM platform was configured to monitor authentication logs from:

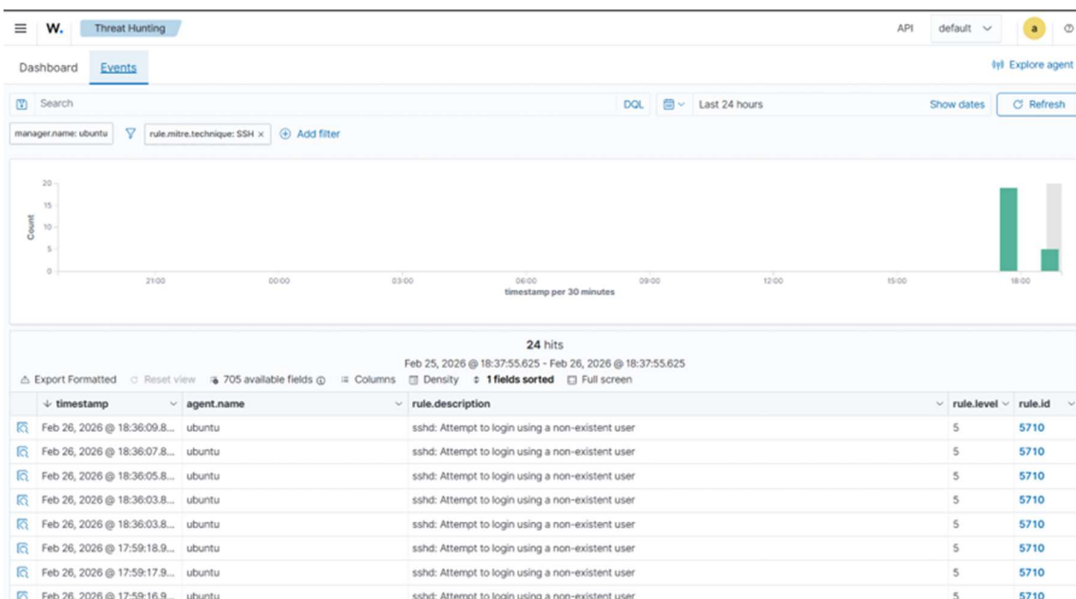
- /var/log/auth.log (Ubuntu Endpoint)

Repeated unauthorized login attempts generated security events which were successfully detected by the built-in Wazuh rule.

The generated alert details are as follows:

- Rule ID: 5710
- Description: sshd: Attempt to login using a non-existent user
- Agent Name: ubuntu
- Severity Level: 5
- Source IP: 192.168.1.9

This indicates multiple unauthorized SSH login attempts which may represent brute-force credential attacks.





5. MITRE ATT&CK Mapping

Detected brute-force activity was mapped to:

- Tactic: Credential Access
- Technique: T1110 – Brute Force

This mapping confirms that repeated authentication failures align with adversarial brute-force techniques commonly used in real-world attacks.

213 hits						
Feb 25, 2026 @ 18:41:31.145 - Feb 26, 2026 @ 18:41:31.145						
Export Formatted Reset view 705 available fields Columns Density 1 fields sorted Full screen						
timestamp	agent.name	rule.mitre.id	rule.mitre.tactic	rule.description	rule.level	rule.id
Feb 26, 2026 @ 18:36:32.5...	Windows-vm	T1484	Defense Evasion, Privilege Escalation	Special privileges assigned to new logon.	3	67028
Feb 26, 2026 @ 18:36:32.5...	Windows-vm	T1078	Defense Evasion, Persistence, Privileg...	Windows Workstation Logon Success	3	60118
Feb 26, 2026 @ 18:36:32.4...	Windows-vm	T1078	Defense Evasion, Persistence, Privileg...	Windows Workstation Logon Success	3	60118
Feb 26, 2026 @ 18:36:09.8...	ubuntu	T1110.001 T1110.001	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
Feb 26, 2026 @ 18:36:07.8...	ubuntu	T1110.001 T1110.001	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
Feb 26, 2026 @ 18:36:05.8...	ubuntu	T1110.001	Credential Access	PAM: User login failed.	5	5503
Feb 26, 2026 @ 18:36:05.8...	ubuntu	T1110.001 T1110.001	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
Feb 26, 2026 @ 18:36:03.8...	ubuntu	T1110.001 T1110.001	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
Feb 26, 2026 @ 18:35:07.8...	ubuntu	T1078	Defense Evasion, Persistence, Privileg...	PAM: Login session opened.	3	5501
Feb 26, 2026 @ 18:33:13.8...	ubuntu	T1548.003	Privilege Escalation, Defense Evasion	First time user executed sudo.	4	5403
Feb 26, 2026 @ 18:33:13.8...	ubuntu	T1078	Defense Evasion, Persistence, Privileg...	PAM: Login session opened.	3	5501
Feb 26, 2026 @ 18:29:03.4...	ubuntu	T1078	Defense Evasion, Persistence, Privileg...	PAM: Login session opened.	3	5501
Feb 26, 2026 @ 18:29:03.2...	ubuntu	T1078	Defense Evasion, Persistence, Privileg...	PAM: Login session opened.	3	5501
Feb 26, 2026 @ 18:28:29.4...	ubuntu	T1548.003	Privilege Escalation, Defense Evasion	Successful sudo to ROOT executed.	3	5402

6. SOC Workflow Implementation

The following SOC workflow was followed during the exercise:

- Detection – Alert triggered in SIEM
- Triage – Verified repeated failed login pattern
- Investigation – Correlated logs and timestamps
- Response – Documented event and monitored source IP
- Reporting – Exported logs and recorded findings

This structured workflow ensures consistent incident handling procedures.

Conclusion

The practical SOC lab successfully demonstrated:

- Log filtering and analysis
- Brute-force attack simulation
- Alert rule creation and validation
- Dashboard visualization
- Browser forensic artifact analysis



- Structured security documentation

The exercises replicate real-world SOC monitoring operations and provide foundational experience in threat detection, investigation, and reporting within a controlled environment. The implementation validates practical understanding of brute-force detection techniques, SIEM alert engineering, dashboard visualization, and SOC workflow execution in a simulated enterprise environment.

References

- Splunk Documentation
- Wazuh Documentation
- MITRE ATT&CK Framework